SDI Investment Property Marketplace

System Documentation

What has been built, how to run it, and who can see what

Repository	`github.com/neilgreene/property-management`
Branch	`main`
Commits	13
Database	PostgreSQL 16 (row-level security; requires 15+)
Application	Node.js 18+ (no framework, no runtime dependencies beyond `pg`)
Automated checks	63 worker tests plus 4 SQL walkthroughs, all passing
Deployment status	Local development only. Not deployed, not internet-facing.

Every fact in this document was extracted from the repository and from a freshly built database, not written from memory. Regenerate it with `python3 docs/generate_system_documentation.py`.

This page intentionally left blank.

Contents

1. What This Is

A working system for a private investment property marketplace: vetted properties listed with financial analysis, browsable without revealing addresses, with the full detail unlocked only after an investor signs and pays the platform fee agreement. Agents see only their own assignments. Staff see everything.

1.1 The problem it solves first

The commercial model depends on one thing holding: an unpaid visitor must not be able to obtain a property's street address. Everything else is ordinary software; that is not. So the visibility rules are enforced **inside the database**, by row-level security policies and column grants, rather than by application code.

> This matters because it changes what an attacker has to defeat. If the rule lives in application code, anyone who finds an unguarded query, an API bug, or a direct database connection gets the address. Here, the address is withheld by the database itself: a caller who writes their own SQL, bypasses the application entirely, and connects directly still cannot read it.

1.2 Three visibility bands

Band	Contains	Released to	Enforced by
1 — Public	City, price, cap rate, NOI, beds, baths, year built	Anyone, including anonymous visitors	RLS policy per role
2 — Gated	Street address, unit, parcel number, seller disclosure, true coordinates	Investors who signed AND paid; the assigned agent; staff	Masking in a `security_invoker` view, keyed on a data predicate
3 — Internal	Acquisition cost, source channel, staff notes, margin	Staff only	Column-level GRANT (a hard ACL)

Coordinates degrade rather than disappear. An ungated viewer gets a deterministic offset of roughly one kilometre, seeded on the property id, so a map still renders a neighbourhood and repeated page loads cannot be averaged to recover the true point.

2. How To Run It

2.1 Docker — nothing installed but Docker

```
git clone https://github.com/neilgreene/property-management
cd property-management
docker compose up
```

Builds the database from `sql/`, seeds it, and serves the demo at **http://localhost:3000**. PostgreSQL is exposed on `localhost:5432` (database `sdi`, user `postgres`, password `postgres`). `docker compose down -v` discards the database.

2.2 Local PostgreSQL 16+ and Node 18+

```
./run.sh
```

Loads all eleven schema files, runs all four SQL walkthroughs, runs the 63 worker tests, then starts the demo. It assumes `psql` and `createdb` work as your own user, which is the default for Postgres.app and Homebrew.

2.3 Individual pieces

Command	What it does
`psql -d sdi -f sql/05_tests.sql`	Security walkthrough: 11 checks, 5 of them attacks
`psql -d sdi -f sql/07_ghl_tests.sql`	GoHighLevel bridge: 7 checks
`psql -d sdi -f sql/10_review_tests.sql`	Review queue actions: 7 checks
`psql -d sdi -f sql/14_pipeline_tests.sql`	Deal visibility and stage history: 9 checks
`cd worker && npm test`	63 unit and end-to-end tests

2.4 The integration worker

Deliberately not started by a bare `docker compose up`, because it needs real GoHighLevel credentials and there is no point running it without them.

```
GHL_TOKEN=... GHL_LOCATION_ID=... docker compose --profile worker up
```

Variable	Where it comes from
`GHL_TOKEN`	GoHighLevel: Settings → Private Integrations. Scoped to an entire sub-account.
`GHL_LOCATION_ID`	The sub-account id, visible in the GoHighLevel URL
`GHL_WEBHOOK_PUBLIC_KEY`	GoHighLevel's published webhook verification key (PEM)

The token is read from the environment and is never written into any file in the repository. It grants access to the whole sub-account — contacts, invoices, transactions — so it must never reach browser-side code.

3. Users and Access

3.1 There is no login system yet

> **This must be stated plainly, because it is the most likely thing to be misread. The system has no authentication: no password for a person, no session, no sign-in page. The demo presents a persona switcher, and choosing a persona makes the web tier assume the matching database role and set that person's id for the duration of one transaction.**

That is deliberate for this stage and costs nothing later. The database contract is identical either way: in production the persona and actor id come out of an authenticated session instead of a dropdown, and not one policy, view or grant changes. Authentication is the missing piece — the authorisation model beneath it is complete and tested.

3.2 Seeded people

Created by `sql/04_seed.sql`. These are demonstration records, not real people; the addresses and financials attached to them are invented.

Name	Role	Email	Fee agreement	Brand	What they demonstrate
Jessica Pool	admin	`jpool@yahoo.com`	n/a	BRAND_A	Full staff access, all bands
Dan Beitor	admin	`dan@example.com`	n/a	BRAND_A	Full staff access, all bands
Tom Bradbury	agent	`tom@example.com`	n/a	BRAND_A	4 assigned properties, incl. an unpublished draft
Priya Raman	agent	`priya@example.com`	n/a	BRAND_A	A second agent, to prove agents are isolated
Ruth Okonkwo	investor	`ruth@example.com`	signed	BRAND_A	Gate open: sees addresses and true coordinates
Marcus Pell	investor	`marcus@example.com`	not signed	BRAND_A	Gate shut: same query, address withheld
Ines Duarte	investor	`ines@example.com`	signed	KAVADOO	The second brand, at concierge pricing

Ruth and Marcus are the pair that carries the argument: identical role, identical SQL, and the only difference between them is one timestamp in a data column. The address appears for one and not the other with no application logic involved at all.

3.3 Database roles

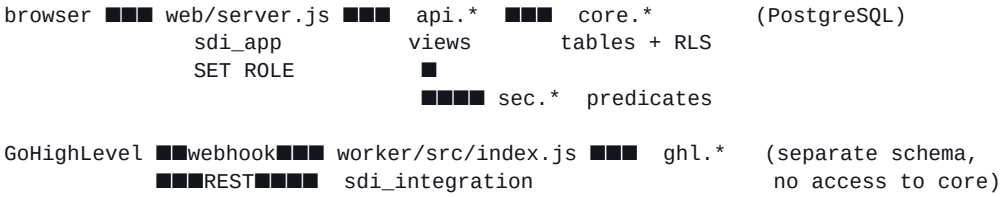
Application roles are created `NOLOGIN` and passwordless on purpose. `sdi_app` is `NOINHERIT`, so it holds no privileges of its own and must assume exactly one persona role per transaction — the application cannot forget to assume a role and accidentally run with more authority than it should.

Role	Purpose	Login	Demo password
`sdi_app`	The only role the web tier connects as	yes	`demo_app_pw`
`sdi_public`	Anonymous visitors. Band 1 only	no	assumed via SET ROLE

Role	Purpose	Login	Demo password
`sdi_investor`	Investors. Band 2 if the gate is open	no	assumed via SET ROLE
`sdi_agent`	Agents. Band 2 on assigned properties	no	assumed via SET ROLE
`sdi_admin`	Staff. All three bands	no	assumed via SET ROLE
`sdi_integration`	The GoHighLevel worker. No access to core at all	yes	`demo_int_pw`
`sdi_test_admin`	Test fixtures only. `BYPASSRLS`	yes	`demo_test_pw`

Those passwords come from `sql/99_local_logins.sql`, exist so the demo stack is connectable, and are published in a public repository. They are worth exactly nothing and that file must not be loaded anywhere that matters. `sdi_test_admin` carries `BYPASSRLS` and belongs only in a test database.

4. Architecture



4.1 Four schemas, and the separation is load-bearing

Schema	Holds	Who has USAGE
`core`	Base tables. Properties, people, deals, assignments	No application role. Name resolution fails before any ACL or policy is consulted, so there is no path around the views.
`sec`	Security predicates, definer-rights	All persona roles. They live here because a `security_invoker` view resolves the functions it calls as the caller too, not just the tables.
`api`	Masking views and the write path	All persona roles. The only surface the application reads.
`ghl`	The GoHighLevel bridge	`sdi_integration` only. The web tier cannot enumerate CRM contacts, invoices or transactions.

Brand is a lens, not a property attribute. Both brands read the same `core.property` rows; `core.property_brand` decides publication and price per brand. Adding the second brand is rows in one table, not a schema refactor.

4.2 Tables

Schema	Tables
`core`	`brand`, `person`, `property`, `property_brand`, `property_assignment`, `saved_property`, `pipeline`, `pipeline_stage`, `deal`, `deal_stage_history`
`ghl`	`id_map`, `webhook_event`, `transaction`, `fee_agreement`, `sync_state`, `outbox`, `review_queue`, `reviewable_field`

All ten `core` tables have row-level security enabled and forced; none of the `ghl` tables do, because that schema is reachable only by the integration worker and by nothing else.

Every column of every table is defined in Appendix A, with its type, nullability, default, key role and foreign key target. That appendix is generated from the live database rather than transcribed, so it cannot describe a schema the system does not actually have.

5. The API

5.1 Database views — what the application reads

View	Returns	Granted to
`api.property`	Properties with band 2 masked or released per caller	public, investor, agent, admin

View	Returns	Granted to
`api.property_internal`	Band 3: acquisition cost, source, notes, margin	admin only
`api.my_saved`	The caller's own saved properties	investor, admin
`api.deal`	Deals the caller is party to, with stage and age	investor, agent, admin
`api.deal_history`	Stage transitions for those deals	investor, agent, admin
`api.review_open`	Inbound CRM edits awaiting a decision	admin only

Note that `api.deal` is not granted to `sdi_public` at all. A deal is never public, at any stage.

5.2 Callable functions

Function	Does	Caller
`api.save_property(uuid)`	Saves a property to the caller's list, rejecting anything they cannot see	investor
`api.review_decide(bigint, text)`	Accept or reject a queued CRM edit; accepting writes an allowlist of columns only	admin
`api.security_invariants()`	Returns zero rows, or names what is broken	admin
`ghl.apply_fee_agreement(text)`	Opens the gate for a person, if the document is completed AND paid	integration worker

The `sec.*` predicates (`can_see_address`, `can_see_deal`, `is_assigned`, `actor`, `jitter` and others) are internal. They are the shared source of truth that policies, views and write functions all consult, so a rule cannot drift between the place it is read and the place it is enforced.

5.3 HTTP endpoints

Service	Method and path	Purpose
Web demo	`GET /`	The demo interface
Web demo	`GET /api/view?persona=&brand=`	Everything one persona sees, in one payload
Web demo	`GET /api/probe?persona=`	Attempts a direct base-table read, to show the refusal
Worker	`POST /webhooks/ghl`	Receives a GoHighLevel delivery, verifies its signature
Worker	`GET /healthz`	Queue depth: pending events, bad signatures, outbox backlog, open reviews

The web demo runs on port 3000, the worker on 3001. Neither is authenticated; neither should be exposed to a network until section 8 is addressed.

6. The GoHighLevel Integration

Built against GoHighLevel's official OpenAPI specifications. The full API contract is documented separately in `docs/GHL-Interface-Specification.pdf` (17 pages).

6.1 What the worker does

Component	Responsibility
`ghlClient.js`	HTTP client. Sets the required `Version: 2021-07-28` header once, paces under the 100-request/10-second limit, retries 429 and 5xx, never retries 403
`signature.js`	Verifies webhook signatures against GoHighLevel's published RSA public key
`webhookReceiver.js`	Records deliveries, deduplicating on `webhookId` and rejecting stale ones
`handlers.js`	Dispatches received events
`outbox.js`	Drains outbound writes with retry that cannot duplicate a record
`sync/documents.js`	Polls fee agreement state
`sync/transactions.js`	Nightly reconciliation of payments
`migrate/`	EspoCRM to GoHighLevel load, in ordered passes

6.2 The fee gate has two conditions, not one

GoHighLevel reports document status and payment status *independently*. A document can be signed and unpaid. `ghl.apply_fee_agreement()` is the only thing in the system that opens the gate, and it requires both: completed or accepted, **and** paid. A tag-based unlock gets this wrong.

6.3 Direction decides what happens to an inbound event

`core.property` is the system of record; GoHighLevel is downstream of it for listings. So an inbound record edit means somebody changed a property inside the CRM, against the grain of the architecture. Applying it would let the CRM silently overwrite the authoritative row. Dropping it would lose a real edit. It is neither: it goes to `ghl.review_queue` for a person to decide.

Events that genuinely originate in GoHighLevel — a signed document, a settled payment, a contact created by staff — are applied directly, because for those GoHighLevel is the source. Accepting a queued edit writes only an allowlist of band 1 columns, so a status change can never become a route to rewriting an address or a cost basis.

6.4 One item is unverified

GoHighLevel publishes the webhook public key but does not document the signature algorithm. The code uses PKCS#1 v1.5 with SHA-256, the conventional pairing for that key format. One captured live delivery would settle it, and until then the receiver should not be trusted in production. `worker/tools/capture-webhook.js` **exists to capture one: run it somewhere GoHighLevel can reach, send a test contact through, and it writes the raw bytes and headers untouched.**

7. What Is Tested

Not a coverage percentage. These are the specific claims that are checked, and the attacks that are run against them.

Suite	Checks	Notable
`sql/05_tests.sql`	11	Five attacks: direct base-table read, another investor's saved list, saving an invisible property, a VOLATILE predicate side-channel, and the standing invariants
`sql/07_ghl_tests.sql`	7	A signed-but-unpaid document must not open the gate; replay must not move a signature timestamp
`sql/10_review_tests.sql`	7	A payload smuggling an address and a cost basis alongside a legitimate status change; only the allowlist is applied
`sql/14_pipeline_tests.sql`	9	One agent cannot read another's deal by naming its id; one investor cannot read another's stage history
`worker/ (npm test)`	63	Signature forgery, replay, oversized bodies, rate-limit handling, ambiguous-create duplication, migration resumability

`api.security_invariants()` must always return zero rows. It catches the four changes that quietly dismantle the model: USAGE granted on `core`, an internal column exposed to a non-admin, RLS switched off on a protected table, or a view created without `security_invoker`. Wire it into CI and a nightly check.

8. What Is Not Built

Stated plainly so nothing here is mistaken for finished.

Missing	Consequence
Authentication	No login, no sessions, no passwords for people. The demo selects a persona. This is the gap between the current system and anything user-facing.
The public marketplace UI	Only the demo interface exists. It exercises the model; it is not the product.
Document storage	The signed PDF artifact is not stored
Messaging and unified inbox	Not started
Audit trail on band 2 and 3 reads	Who viewed which address, and when, is not recorded
Co-investment matching	The pipeline exists; the matching engine does not
The external status scraper	Not built. The review queue that would receive its output is.
EspoCRM field mapping	The load ordering and resumability are built and tested. The field-level mapping needs the live EspoCRM schema.
Any deployment	Nothing is hosted. Nothing is internet-facing.

9. Where Things Are

Path	Contents
`sql/01–04`	Schema, RLS policies, masking views, demo data
`sql/05, 07, 10, 14`	The four walkthroughs. Each is readable top to bottom as an argument
`sql/06, 08, 09`	GoHighLevel bridge, review queue, review actions
`sql/11–13`	Deals, stage history, pipeline policies and seed
`sql/99_local_logins.sql`	Demo passwords. Local development only
`web/`	The demo: `server.js` and one HTML page
`worker/src/`	The GoHighLevel integration worker
`worker/test/`	63 tests
`worker/tools/`	The webhook capture tool
`docs/`	This document and the GoHighLevel interface specification, with their generators
`README.md`	The same ground in more technical detail, including the reasoning behind each design decision

There is no INSTALL file; section 2 of this document and the README's opening section cover installation. Both PDFs in `docs/` *are generated by committed scripts rather than written by hand, so they can be regenerated as the system changes and cannot quietly drift out of date.*

Appendix A. Table Definitions

Every column of every base table, read from the live database by `docs/extract_schema.py` and stored in `docs/schema-snapshot.json`. Regenerate the snapshot after any schema change and rebuild this document; the appendix cannot then describe a schema the system does not have.

Marker	Meaning
PK	Part of the primary key
FK	Foreign key; the referenced table is named in the same row
UQ	Covered by a unique constraint
CK	Covered by a check constraint
not null	The column is `NOT NULL`

Schema `core`

`core.brand`

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
`brand_code`	`text`	not null	PK	
`display_name`	`text`	not null		
`service_tier`	`text`	not null	CK	
`platform_fee`	`numeric(10,2)`	not null		

`core.deal`

13 columns · row-level security enforced

Column	Type	Null	Key	Default / references
`deal_id`	`uuid`	not null	PK	gen_random_uuid()
`external_ref`	`text`		UQ	
`property_id`	`uuid`	not null	FK	→ core.property
`investor_id`	`uuid`		FK	→ core.person
`agent_id`	`uuid`		FK	→ core.person
`pipeline_code`	`text`	not null	FK	→ core.pipeline_stage
`stage_code`	`text`	not null	FK	→ core.pipeline_stage
`amount`	`numeric(12,2)`			
`opened_at`	`timestamptz`	not null	CK	now()
`closed_at`	`timestamptz`		CK	
`lost_reason`	`text`			
`created_at`	`timestamptz`	not null		now()

Column	Type	Null	Key	Default / references
updated_at	timestamptz	not null		now()

core.deal_stage_history

7 columns · row-level security enforced · Append-only, written by trigger. The application cannot forget to log a transition, and cannot rewrite one after the fact.

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment
deal_id	uuid	not null	FK	→ core.deal
from_stage	text			
to_stage	text	not null		
changed_at	timestamptz	not null		now()
changed_by	uuid			
seconds_in_from	bigint			

core.person

7 columns · row-level security enforced

Column	Type	Null	Key	Default / references
person_id	uuid	not null	PK	gen_random_uuid()
role	actor_role	not null		
full_name	text	not null		
email	text	not null	UQ	
fee_agreement_signed_at	timestamptz			
home_brand	text		FK	→ core.brand
active	boolean	not null		true

core.pipeline

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
pipeline_code	text	not null	PK	
display_name	text	not null		
brand_code	text		FK	→ core.brand
active	boolean	not null		true

core.pipeline_stage

6 columns · row-level security enforced

Column	Type	Null	Key	Default / references
pipeline_code	text	not null	FK PK UQ	→ core.pipeline
stage_code	text	not null	PK	
display_name	text	not null		
position	integer	not null	UQ	
is_won	boolean	not null	CK	false
is_lost	boolean	not null	CK	false

core.property

27 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	PK	gen_random_uuid()
listing_ref	text	not null	UQ	
status	text	not null	CK	
city	text	not null		
state	char(2)	not null		
zip	text	not null		
property_type	text	not null		
beds	smallint			
baths	numeric(3,1)			
sqft	integer			
year_built	smallint			
list_price	numeric(12,2)	not null		
gross_rent_annual	numeric(12,2)	not null		
opex_annual	numeric(12,2)	not null		
hoa_annual	numeric(12,2)	not null		0
noi_annual	numeric(12,2)			
cap_rate	numeric(6,4)			
street_address	text	not null		
unit	text			
lat	numeric(9,6)	not null		
lng	numeric(9,6)	not null		
parcel_number	text			
seller_disclosure	text			
acquisition_cost	numeric(12,2)			

Column	Type	Null	Key	Default / references
source_channel	text			
internal_notes	text			
created_at	timestamptz	not null		now()

core.property_assignment

3 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
person_id	uuid	not null	FK PK	→ core.person
assign_role	text	not null	CK PK	

core.property_brand

4 columns · row-level security enforced

Column	Type	Null	Key	Default / references
property_id	uuid	not null	FK PK	→ core.property
brand_code	text	not null	FK PK	→ core.brand
published	boolean	not null		false
brand_price	numeric(12,2)			

core.saved_property

3 columns · row-level security enforced

Column	Type	Null	Key	Default / references
person_id	uuid	not null	FK PK	→ core.person
property_id	uuid	not null	FK PK	→ core.property
saved_at	timestamptz	not null		now()

Schema ghl

ghl.fee_agreement

10 columns · no row-level security

Column	Type	Null	Key	Default / references
document_id	text	not null	PK	
location_id	text	not null		
person_id	uuid		FK	→ core.person
ghl_contact_id	text			
status	text	not null		
payment_status	text	not null		

Column	Type	Null	Key	Default / references
grand_total	numeric(12,2)			
ghl_updated_at	timestamptz	not null		
observed_at	timestamptz	not null		now()
raw	jsonb	not null		

ghl.id_map

6 columns · no row-level security

Column	Type	Null	Key	Default / references
entity_type	text	not null	PK	
local_id	uuid	not null	PK	
ghl_id	text	not null	UQ	
ghl_object	object_kind	not null		
location_id	text	not null	PK UQ	
synced_at	timestamptz	not null		now()

ghl.outbox

13 columns · no row-level security

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment
idempotency_key	text	not null	UQ	
operation	text	not null		
entity_type	text			
local_id	uuid			
payload	jsonb	not null		
state	outbox_state	not null		'pending'
attempts	integer	not null		0
next_attempt_at	timestamptz	not null		now()
last_error	text			
ghl_id	text			
created_at	timestamptz	not null		now()
sent_at	timestamptz			

ghl.review_queue

12 columns · no row-level security

Column	Type	Null	Key	Default / references
id	bigint	not null	PK	auto-increment

Column	Type	Null	Key	Default / references
source	text	not null		
event_type	text	not null		
ghl_object	text			
ghl_id	text			
local_id	uuid			
summary	text	not null		
proposed	jsonb	not null		
state	review_state	not null		'open'
raised_at	timestamptz	not null		now()
decided_at	timestamptz			
decided_by	uuid			

ghl.reviewable_field

1 columns · no row-level security · Allowlist. An accepted CRM edit can write these and nothing else. Deliberately excludes every band 2 and band 3 column: a status change must not be a route to rewriting an address or a cost basis.

Column	Type	Null	Key	Default / references
column_name	text	not null	PK	

ghl.sync_state

7 columns · no row-level security

Column	Type	Null	Key	Default / references
resource	text	not null	PK	
location_id	text	not null		
cursor_at	timestamptz			
last_run_at	timestamptz			
last_ok_at	timestamptz			
last_error	text			
records_seen	bigint	not null		0

ghl.transaction

17 columns · no row-level security

Column	Type	Null	Key	Default / references
ghl_id	text	not null	PK	
location_id	text	not null		
contact_id	text			
invoice_id	text			

Column	Type	Null	Key	Default / references
subscription_id	text			
amount	numeric(12,2)	not null	CK	
currency	char(3)	not null		
amount_refunded	numeric(12,2)	not null	CK	0
status	text	not null		
live_mode	boolean	not null		
payment_provider	text			
entity_type	text			
entity_id	text			
ghl_created_at	timestamptz	not null		
ghl_updated_at	timestamptz	not null		
synced_at	timestamptz	not null		now()
raw	jsonb	not null		

ghl.webhook_event

9 columns · no row-level security

Column	Type	Null	Key	Default / references
webhook_id	text	not null	PK	
event_type	text	not null		
occurred_at	timestamptz	not null		
received_at	timestamptz	not null		now()
processed_at	timestamptz			
attempts	integer	not null		0
last_error	text			
signature_ok	boolean	not null		
payload	jsonb	not null		